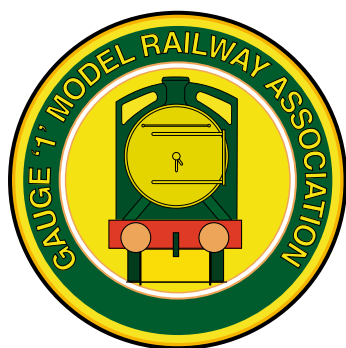




Modelling in Gauge 1



Book 3 : Freight stock

Published by G1MRA 2007

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The Gauge 1 Model Railway Association’s principle objectives are to promote railway modelling in a scale which will involve a rail gauge of approximately 1.75 inches or 45 millimetres, and to bring together persons interested in the construction of Gauge One models, tools and apparatus of all kinds.

Information on G1MRA can be obtained from their web site at www.gaugeone.org or by writing to the Membership Secretary at PO Box 581, Northampton, NN6 0YW, England.

The upper picture on the front cover is the 10 ton Private Owner's wagon, as made by Gloucester Carriage and Wagon Works in 1901 for Crynant Colliery Company. It has springs from silicone rubber on piano wire, with doors and brake fully functional. This won a Gold Medal for Roy Froom at the Model Engineers Exhibition in 1994.
See the article by Roy Froom (Autumn 2002 195/18) (Photo by Roy Froom.)

The lower picture on the front cover is a steam roller truck No 41000 to Dia F2 carrying its allocated Aveling & Porter road steam roller
See the article by David West Spring 2002 193/16 59 (Photo by Dave West)

The upper picture overleaf shows a scratch built 5 ton travelling hand operated crane
See the article by Trefor Wheeler Autumn 2001 191/24 (Photo by Trefor Wheeler)

The lower picture overleaf shows a finished wagon kit
See the article by David Stick Spring 2003 197/7 (Photo by David Stick)

Introduction to Modelling in Gauge 1 Robert Houghton - President of G1MRA

Some 80 years ago I used to look forward to the next monthly issue of the then Model Railway News. Now, in my 96th year, I still have that same anticipation, but for the next issue of the Gauge One Newsletter and Journal, which never fails to come up to expectations. Could this be my second childhood? Not a bit of it; I just had the good fortune all those years ago to become interested in the Gauge One Model Railway hobby, and it has remained with me for the rest of my long life, with the added incentive of being able to share it with my two sons, both of whom have garden railways of their own. Age does not come into it. For the younger generations it is a hobby which provides a constructive and positive alternative to the TV and computer screens, whilst for those in retirement I can recommend it as an aid to keeping an active mind; and you can always borrow a magnifying glass! Gauge One is such a wide-ranging hobby, for young or old, steam or electric, indoors or out in the garden, skilled or unskilled, and above all as a means of meeting all manner of interesting and friendly people you would otherwise never have heard of.

Since the earliest days of our Association, in 1948, the Newsletter has been a central feature. Regularly produced and edited by volunteers, it has contained above all a wealth of contributions by the more skilled, erudite and experienced Members for the benefit of the rest of us. There is always something new to be learnt, and encouragement to be obtained from the descriptions of other people’s ideas and accomplishments. Typical of the spirit of G1MRA, Geoffrey Hammond and his team of helpers have voluntarily delved into over 200 editions of the Newsletter, and undertaken the laborious process of compiling and publishing this, the third of a new series of booklets. Our sincere thanks go to them for this further contribution to our enjoyable and fascinating Gauge One hobby.

I am delighted to have this opportunity to express my personal thanks for the help I have obtained from the Newsletters over the years. These new booklets, I am sure, are going to be a source of encouragement and enjoyment, particularly to newcomers to our hobby. In the words of one of our more illustrious Members, the late Freddie Wrighton: “have a go at it”. You may be pleasantly surprised by what you can achieve.

It was my Doctor, Dr. Greenway, who showed me Gauge 1 while I was still at school and he encouraged me to build some coaches. Shortly after this I visited the Model Railway Club Easter Exhibition and from the Gauge 1 stand I obtained a membership application and purchased a copy of “Modelling in Gauge 1”. This booklet published in 1972 contained a selection of useful articles that had appeared in Newsletters 1 to 83. Over the following years many more useful and informative articles and letters have been written and there have been calls for an updated Modelling in Gauge 1. In response to those calls this is the third of a planned series of books drawing on the many articles and letters that have been published over more than 50 years of the Association from over 200 Newsletters. These books have been in the planning and production stage for 12 years and progress has been made as time and commitments have allowed. At times some have thought the proposal moribund.

We have drawn on the help and experience of various people to compile these books and for the third book our grateful thanks are offered to Peter Howland. Peter was rightly concerned that some references to suppliers in today’s context have little relevance, so we have included an appendix giving current suppliers for freight stock items at the time of publishing.

Additionally, our thanks must go to the many contributors over the years, without whom the newsletter would be just blank pages; also the various newsletter Editors – Ella Roberts/Price, Kevin West, Martin & Pauline Hulse, Art Walker and Dick Comber; and to Martin Hulse for production editing, including typesetting and layout, of this book.

It has not been our intention to rewrite these articles but present them in the same hand that the Author originally prepared them, so we have done the minimum of editing that was felt necessary to make the articles readable. We have corrected obvious errors of spelling and punctuation. We have added comments or updates where these add to the article. We have removed opening and closing salutations, and invitations to future meetings. Over the years the Newsletter (latterly the Newsletter and Journal) has changed format several times. We have chosen to use a similar layout to the current Newsletter and Journal format for this book.

Articles are published in this book in their original Newsletter issue order, with the oldest first. Each article shows in its heading the title, author, year originally published, issue number, and page number within that issue. The contents section on page 5 lists these headings, together with the page number within this book where they appear. So for example:-

SPRINGTIME ? George Thompson Mar 1978 100/9
appeared on page 9 of Newsletter Issue 100 which was published originally in March 1978. The index at the back of this book shows where selected topics appear.

These books hopefully will stimulate their own debate and development so if you feel you can add to this please submit your article to the current Newsletter Editor, assuming you are a G1MRA member, since the G1MRA Newsletter has always been for members only. If you are not a G1MRA member and would like more of the type of articles in this book then join G1MRA and you will receive the Newsletter once a quarter, packed with contributions on a wide variety of Gauge 1 topics, and will also be able to contribute your own views and experiences.

Freight Stock - An introduction Peter Howland

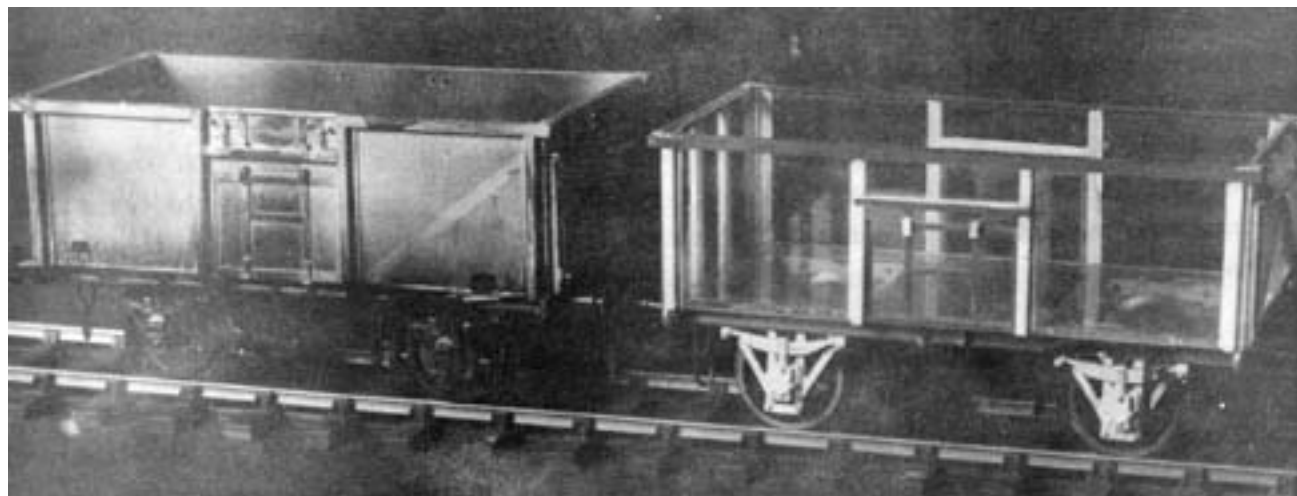
The early railway companies and the big 4 groups had to develop a very extensive range of vehicles to handle the immense variety of traffic on offer. Everything from the open goods and mineral wagons to specialised vehicles for fish, live animals, household removals through to heavy machinery. These vehicles came in many sizes and almost as many liveries.

The articles selected for this book will hopefully encourage the modeller to try his hand at construction of freight stock, and show that Gauge One allows the inclusion of detail such as opening doors and working brake gear.

Due to the lengthy timescale covered by this publication some of the suppliers and products mentioned here may not now be available and alternatives will need to be found. The Gauge One Model Railway Association provides a list of current suppliers to its members, and many of these advertise in the G1MRA Newsletter & Journal. An extract from this list is appended at the end of this book.

A beginner who finds himself in trouble for lack of information will need to look further afield, for example the various modelling and historic publications. Alternatively if they join the Gauge One Model Railway Association they will be able to draw on the resources of other members, many of whom have already explored their specific chosen interests and accumulated relevant information.

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It soon became apparent when I started on my first Gauge 1 layout, a colliery set in the late fifties, that I was going to need more than a few 16ton mineral wagons. Thousands of these were built in real life and nearly every photo from that era seems to have at least a couple somewhere. In every other scale of model railways one can buy both kit and ready to run ones but in Gauge 1 one has to scratchbuild what is quite a complicated vehicle.

The first job was to prepare a 10mm drawing and I used as a basis one of Ken Werretts from a back number of Model Railway News. I also photographed some of the real ones albeit fitted ones, unfitted wagons now virtually extinct except for some engineers stock. The second task was to decide what material to construct them from. Wood was out for the obvious reasons and the thought of two dozen made from metal didn't thrill me at all! So it looked like being Plasticard despite its tendency to crumble with age and warp and its somewhat temperamental relationship with some paints. However just before I started Fate took a hand.

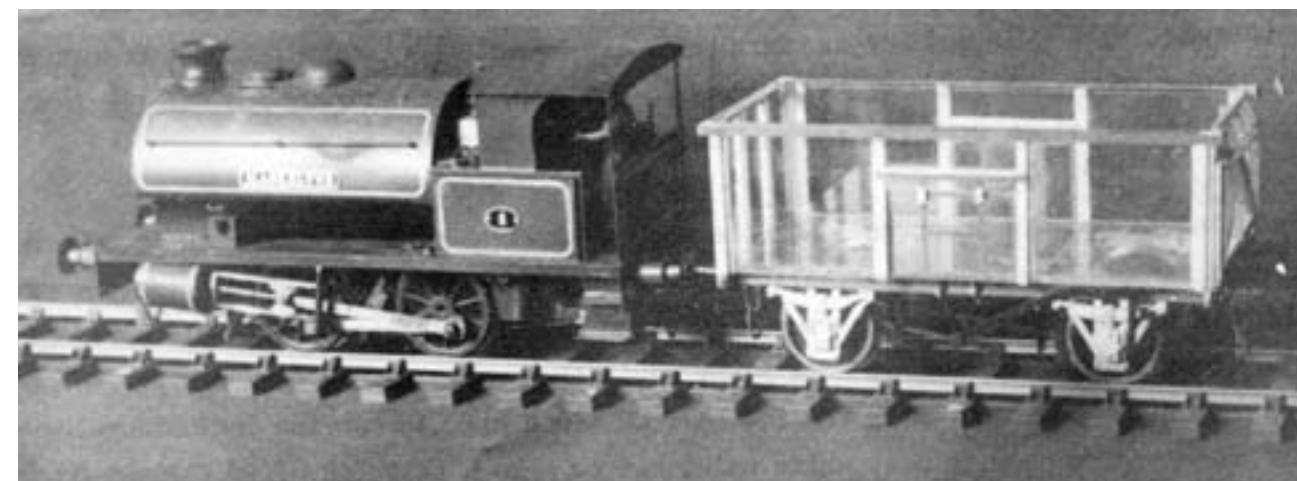
For the last few years I have toured with an 0 gauge layout called Houghton, indeed we've clocked up 28 shows with it at the latest count and whilst unloading into a friends house after a show I noticed a pile of clear plastic offcuts under his kitchen sink. It transpired that it was offcuts from his double glazing, a clear sunlight resistant acrylic sheet available from DIY shops under various trade names. In my 0 gauge days I had made some parcel vans and these had proved quite durable in service but they were made from Perspex, ICI's acrylic and that is a lot more expensive. Anyway, after some brief negotiations I came home with a big pile of the stuff.

A brief experiment showed that by scoring it both sides a clean snap could be achieved and that Plastic Weld, the liquid cement sold for use with Plastruct produced a very strong joint, and that conversely Mek Pak didn't touch it at all.

The first box was built and left to set overnight and the following day a start was made on the detailing. The reinforcing angle around the top was made from Plastruct 1/8" angle, Plastruct being a range of extruded ABS sections

available from decent model shops. The corners were mitred and the L shaped reinforcing plates added from 10thou Plasticard. The dressed steel stanchions down the sides were the next problem, initial attempts to cut it from the acrylic sheet were unsuccessful but a military modelling friend, a field I vacated years ago, came up with a solution. Tamiya, the kit people, make a range of polystyrene rodding in 2, 3 and 5mm square. The 3mm rod was notched to fit under the top rail and cut to finish flush with the bottom of the sides. The flanges down the side of the stanchions were added from strips of 20thou plasticard about 1mm wide. At the non door end the stanchions protrude down to reinforce the buffer beam. At the door end it gets a bit complicated, and it helps to look at the real thing. The door frame was made from 2mm square Tamiya rodding and the stiffeners were cut from 40thou Plastruct sheet, the little angle at the top was the smallest in the Plastruct rack. The main side door and top doors framing was again cut from 20thou Plasticard with little chunks of 2mm square to represent the latches. 1mm round plastic rod was adzed down to semicircular section and used for the hinges and the rail along the top of the top door. This same rod was also used for the pins that fix the retaining chains on the top door, by holding the rod near something hot i.e. a soldering iron one can form a neat button on the end.

The under frame channel is 1/4" I section beam from Plastruct although brass could have been used and screwed to the body. Buffer beams are 80thou plasticard drilled and shaped before fixing as the body fixing plates under the centre stanchions. Offcuts of the 80thou were also used to reinforce the joint between the buffer beam and floor. Two cross members were fixed between the solebars sitting between the brake hangers and brake safety straps. From now on its fairly conventional, axleguards are either Association or Tenmille 'oil box' ones as are the wheels. Brakegear, buffers and couplings are Tenmille but note that the underframe is shallower than normal and you will have to cut down the brakegear and cut the bases of the buffer bodies flush with the top and bottom of the bufferbeam.

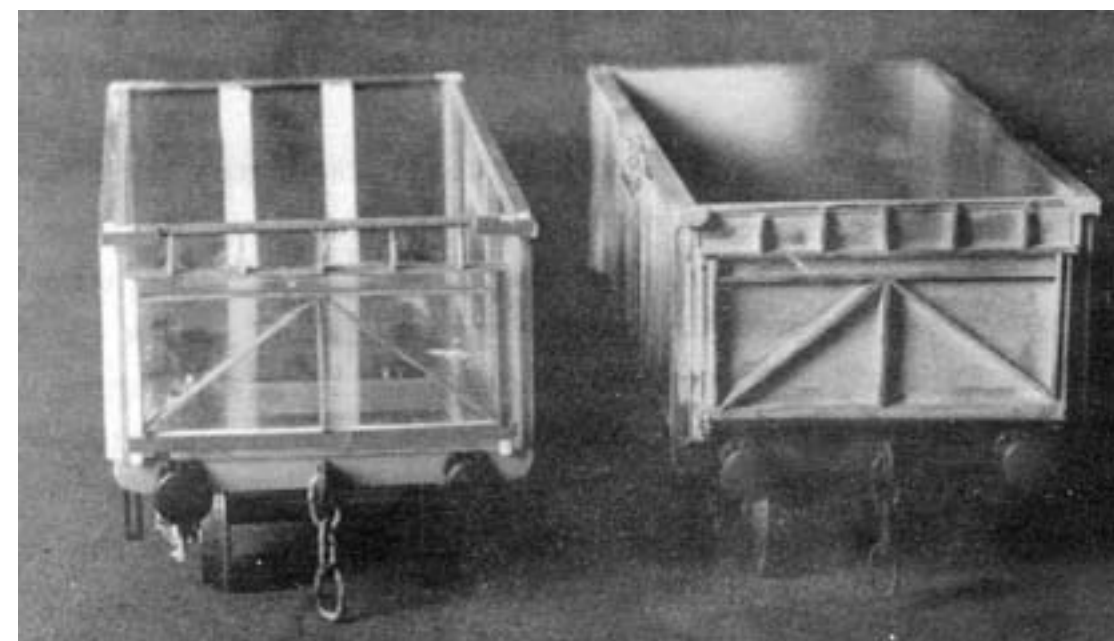


Painting is equally unconventional. The wagon was first given a good wash to remove any grease and perhaps more importantly, any static! Otherwise they seem to attract every bit of dust in the workshop. The whole wagon was then given two coats of matt black. This is to stop the light shining through it on a sunny day! I then use a simple card mask held under the bottom of the body and spray the top half BR wagon grey. The stripe-indicating the end door is hand painted on with matt white paint but to make a neat job of it I first do the edges with a mapping pen and ruler and white indian ink, filling in after with a small brush. Rust is then dry brushed on, and then the wagon subjected to various degrees of weathering. The most effective method seems to be to lightly spray the wagon with matt black with a touch of brown added, whilst still wet spray with thinners so that the black runs down the sides and gets denser at the bottom. Still wet, lightly dust with tan or light brown, spraying heavier around the underframe. Attack the inside of the body with rust and black.

I have made about ten of these wagons now and have

found that batch building 4 or 5 at a time is much more effective, particularly cutting out the main body 'bits' from the sheet of acrylic. As this costs about £6 for a big sheet the total building cost, using Tenmille wheels is about £9 per wagon. I proved the toughness of the stuff when a nearly completed wagon gently rolled off the workbench onto the concrete floor! The only thing to suffer was the brake levers and ratchets and they soon went back on with superglue.

Other potential uses for the stuff includes parcel vans, mask out where the windows are right at the start, scribe door lines etc, build, paint and if appropriate weather the whole vehicle and right at the end peel off the masking. Hey presto! Flush glazing!



The Mike Watts and Kevin West enquiries in the last Newsletter have prompted me to write this description of my latest models.

The first model is of one of the Southern Railway's 25 ton bogie brake vans built in 1936. The photographs show the van which is waiting to be painted.

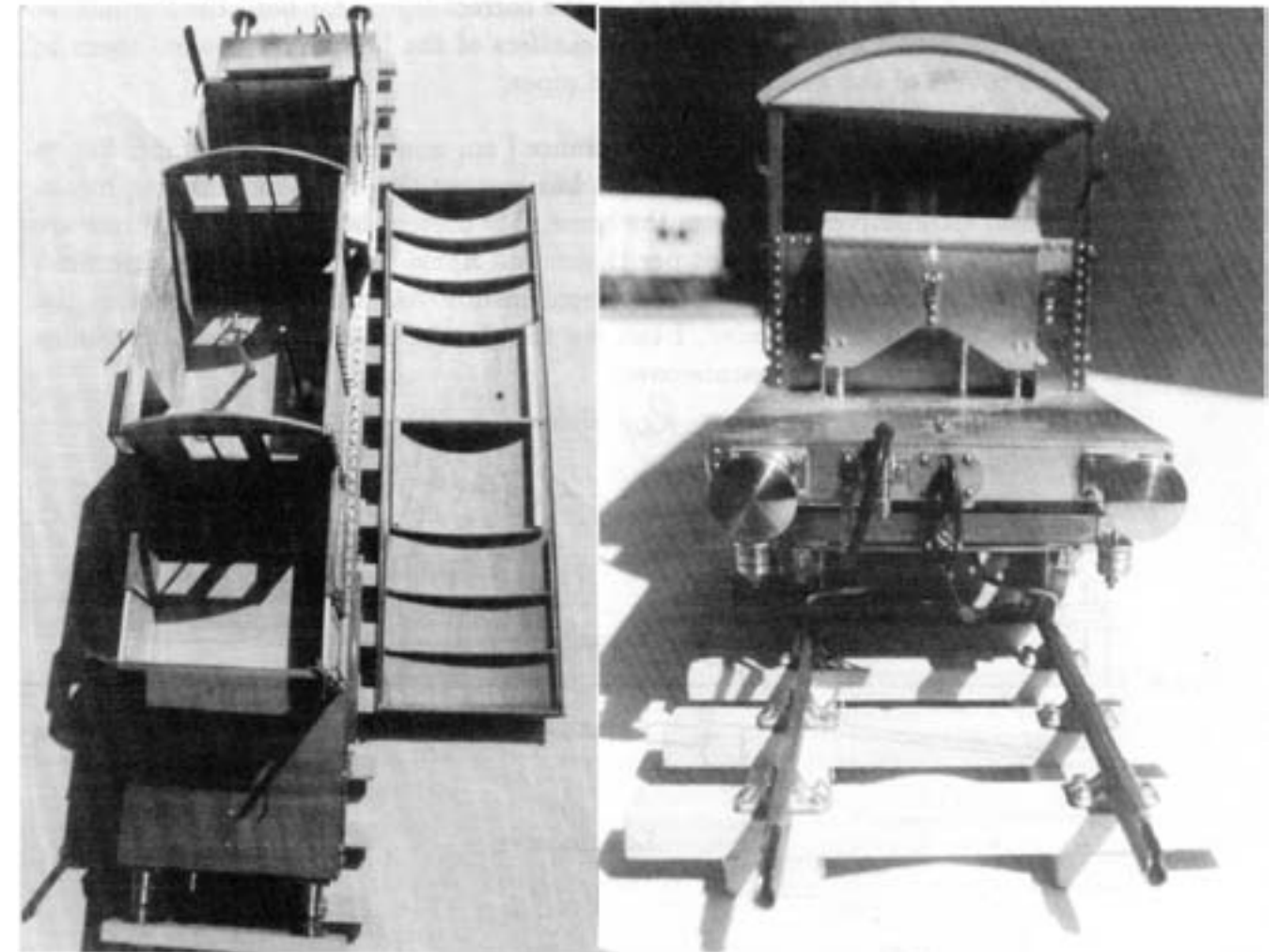
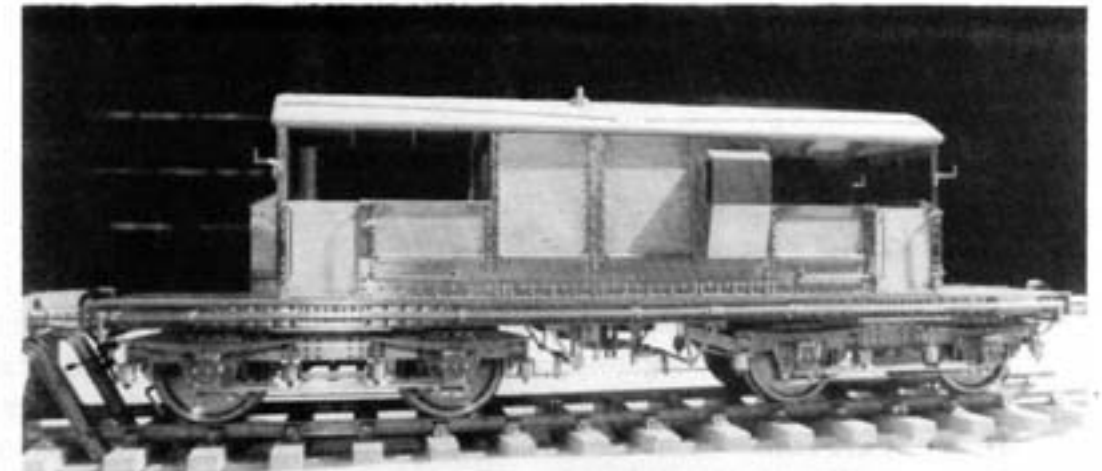
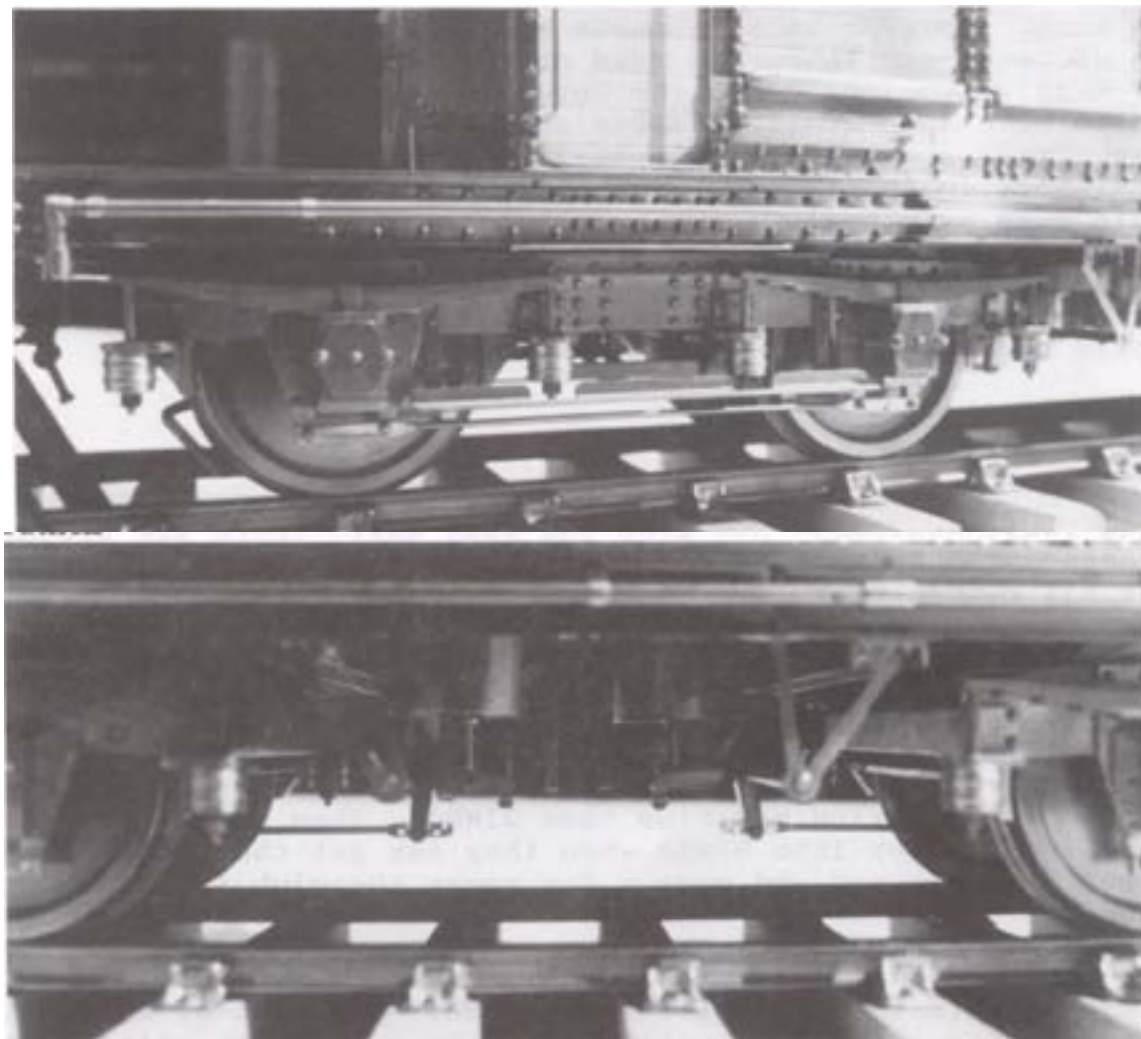
Construction started last year with the bogies. These are a shallow brass tray with the horn guides, spring hangers and other detail riveted on. All wheels are sprung as are the bolsters. Four bogies were made in one production run.

The underframe is made from brass channel soldered to a sheet of brass which forms the decking. The gap between the bogies looked a bit empty and so brake cylinders, vee hangers and brake rodding were added.

The body is made from brass angle, nickel silver sheet and plywood. Some of the rivets hold the layers together but the rest are dummies. Sand boxes on the verandah ends are solid blocks of aluminium sitting on brass angle and channel. The inside of the van is fitted with brake stand, stove coal bin and seats in the lookouts. The roof is removable to enable me to glaze the windows after painting.

Drawings for the van come from the Model Railway Constructor for December 1966. The magazine has a drawing of a 28 ton passenger brake van and the second pair of bogies has been used in the construction of this vehicle.

Most of the brass channel and angle used on these models came from John Flack as did the nickel silver. I ran both vans recently and despite being on the heavy side, they are very free running.



Jib

I decided to make the jib out of plasticard as I didn't fancy trying to solder little bits of brass together. To make the sides of the jib I first made a template out of 14g aluminium plate. Two pieces of twenty thou plasticard were then clamped between the template and a piece of wood. The holes, of various sizes, were then drilled and a very sharp scalpel was used to cut around the template. That was the easy bit. Glueing, with MEK, all the web edges and cross braces to the sides took some time. Once glued together the jib was quite rigid. A brass jib would be more suitable if the jib was to be put to a lot of use.

Supporting the crane jib are two longitudinal arms. These arms retract into rectangular section brass extrusions which are attached to the rear spacing rod between the gear housing sides. There are four holes in the arms which allow the jib to be held at four different heights. The lower height is for travelling and the other three for lifting.

Platforms

Either side of the gear housing are platforms on which the crane operators stood. The platforms were made from thin plywood and very small hinges were used to allow the platforms to fold up for travelling.

I am very pleased with the final result of the unit, crane and match truck. The photos compare the model with the prototype. In the beginning I painted the match truck GWR grey but the unit has now been painted smoky black, Humbrol 85.



To complete the breakdown set I am going to make a mess/tool van. This will be based on a modified GWR Fruit D van and painted brown with red ends. The van will contain extra water and meths for my Aster Pannier, enabling it to run at least thirty minutes before refueling.

When this will be completed I don't know; probably after I've finished the dolls' house, finished the track in the garden and completed all the other little jobs that need doing about the house. How I had time to go to work, before I was made redundant, I don't know.



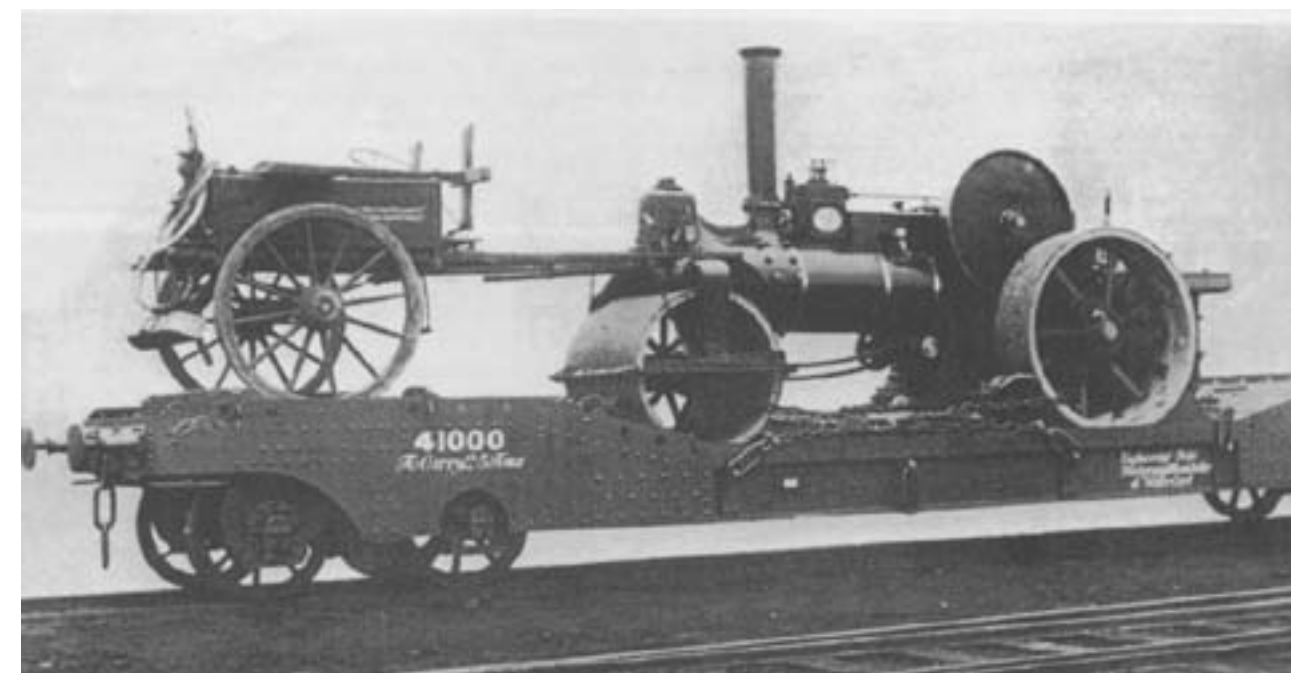
Above: the model. Below: the prototype



A wagon for my steamroller

David West

Spring 2002 193/16



Steam roller truck No 41000 to Dia F2 shown carrying its allocated Aveling & Porter road steam roller 3684/96 and attendant water cart with sprinkler at the rear.

From Atkins, Beard & Tourret, "GW Goods Wagons", Tourret Publishing

The Great Western Railway owned a series of steamrollers, assigned to the Engineering Department, which they used for duties in goods yards and station yards. The first engine, a 12 ton Aveling & Porter No 3684/96, GW No PMW 12, was acquired in 1896, and in 1897 a 25 ton bogie well wagon was built to carry the engine, No 41000.

My interest in the subject of rollers was aroused by my nephew who had just acquired an ex Essex County Council Aveling roller. Thumbing through a copy of Railway Modeller one wet afternoon when outdoor railway activity had come to an abrupt halt, I came across an advertisement for South Eastern Finecast listing, amongst other kits, a 1/32 scale steam roller. A phone call and flash of the credit card soon had me in possession of a kit for an Aveling & Barford steam roller.

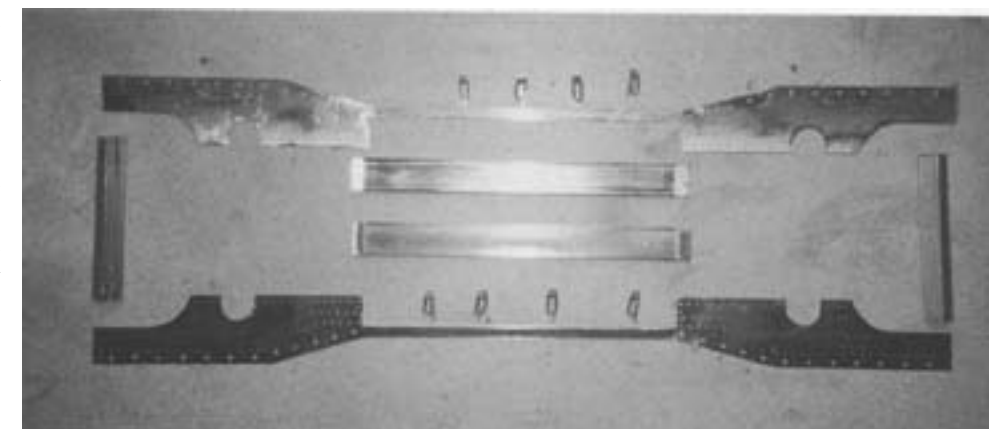
My first idea was to complete the kit and present it to my nephew, but further investigation revealed that the kit was a 10 ton single cylinder model whereas his was a twin cylinder compound. My Christmas present last year was a copy of GWR Goods Wagons, a very comprehensive volume of drawings, photographs and text on the subject including a section on steam roller trucks. That decided it; my nephew was not getting the model. I would build the wagon shown in the photograph to carry the miniature roller.

Fortunately a general

arrangement drawing is included in the book, so my thoughts turned to construction methods. The thing that strikes one about this vehicle is its massive construction and the number of rivets. How were we going to represent all these rivets on the model? Two alternatives came to mind: use thin brass sheet and a rivet press or use thicker brass sheet and real rivets. I decided heavier gauge sheet would be more appropriate. For the main sides and floor 20g brass sheet was the material of choice; the longitudinal beams I fabricated using more 20g sheet cut to width with brass L section soldered along both edges.

Armed with a magnifying glass and looking closely at the photograph, I determined that two sizes of rivet were used. Now I had to count them! – 388 small and 80 larger, yes I counted again just to be sure. The larger rivets were used around the ends of the longitudinal beams. For these 3/64 inch looked about right and for the smaller ones 1/32 inch.

Next problem, rivet spacing. I drew a side of the wagon



The finished frames with main beams and buffer beams